Materials Design Analysis Reporting (MDAR) Checklist for Authors

The MDAR framework establishes a minimum set of requirements in transparent reporting applicable to studies in the life sciences (see Statement of Task: doi:10.31222/osf.oio/9sm4x.). The MDAR checklist is a tool for authors, editors and others seeking to adopt the MDAR framework for transparent reporting in manuscripts and other outputs. Please refer to the MDAR Elaboration Document for additional context for the MDAR framework.

Materials

Antibodies	Yes (indicate where provided: page no/section/legend)	n/a
For commercial reagents, provide supplier name, catalogue number and RRID, if available	No antibodes detected. Please add identifiers for all resources where possible	

Cell Materials	Yes (indicate where provided: page no/section/legend)	n/a
Cell lines: Provide species information, strain. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No cell lines detected Please add identifiers for all resources where possible	
Primary cultures: Provide species, strain, sex of origin, genetic modification status.	Not currently checked by SciScore	

Experimental Animals	Yes (indicate where provided: page no/section/legend)	n/a
Laboratory animals: Provide species, strain, sex, age, genetic modification status. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No organisms detected Please add identifiers for all resources where possible	
Animal observed in or captured from the field: Provide species, sex and age where possible	Not currently checked by SciScore	
Model organisms: Provide Accession number in repository (where relevant) OR RRID	See laboratory animals section for information.	

Plants and microbes	Yes (indicate where provided: page no/section/legend)	n/a
Plants: provide species and strain, unique accession number if available, and source (including location for collected wild specimens)	Not currently checked by SciScore	
Microbes: provide species and strain, unique accession number if available, and source	Not currently checked by SciScore	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
Identify authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Written informed consent was acquired from all participants and the study protocol was approved by the Institutional Review Board of the Samsung Medical Center (IRB No. 2018-02-050) 2.3 Randomization Consecutively recruited participants were randomly assigned to the intervention or control group.	
Provide statement confirming informed consent obtained from study participants.	Written informed consent was acquired from all participants and the study protocol was approved by the Institutional Review Board of the Samsung Medical Center (IRB No. 2018-02-050) 2.3 Randomization Consecutively recruited participants were randomly assigned to the intervention or control group.	
Report on age and sex for all study participants.	**Age:**The following inclusion criteria were applied: (1) individuals **Age:**≥**Age:**60 years old; (2) literate with education years of 6 or higher; (3) Korean version of Mini-Mental State Examination (K-MMSE) score >23 10 ; (4) preserved activities of daily living (ADL), as defined by Korea Instrumental ADL **Age:**≤**Age:**5 11 ; (5) no major neurological and psychiatric disorders affecting cognition or	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
	diagnosis of dementia; and (6) personal Android smartphone users for the purpose of using cognitive training application. **Sex:**not detected.	

Design

Study protocol	Yes (indicate where provided: page no/section/legend)	n/a
For clinical trials, provide the trial registration number OR cite DOI in manuscript.	NCT03494699	

Laboratory protocol	Yes (indicate where provided: page no/section/legend)	n/a
Provide DOI or other citation details if detailed step-by-step protocols are available.	Not detected.	

Experimental study design (statistics details)	Yes (indicate where provided: page no/section/legend)	n/a
State whether and how the following have been done, or if they were not carried out		
Sample size determination	Under the hypothesis that the standard deviation of scores (at baseline or 12 months and in the control or intervention group) was 15 , and the difference between changes of total score (from baseline to 12 month) of the two groups was five , 142 participants were required for each study group with a significance level set at 0.05 and with a power of 80 % to detect a significant difference.	
Randomization	2 METHODS 2.1 Study design This Application-based Cognitive Training at Home (ACTH) study was a single-center, randomized, single-masked, and parallel group study with the aim to investigate the effect of smartphone application-based cognitive training on cognition in community-dwelling elderly individuals.	
Blinding	Cognition was assessed by two independent psychometricians who were blinded to participants' interventions, using a tablet-based cognitive test battery called Inbrain Cognitive Screening Test (	
inclusion/exclusion criteria	The following inclusion criteria were applied: (1) individuals ≥60 years old; (2) literate with education years of 6 or higher; (3) Korean version of Mini-Mental State Examination (K-MMSE) score >23 10 ; (4) preserved activities of daily living (ADL), as defined by Korea Instrumental ADL ≤5 11 ; (5) no major neurological and psychiatric disorders affecting cognition or diagnosis of dementia; and (6) personal Android smartphone users for the purpose of using cognitive training application.	

Sample definition and in-laboratory replication	Yes (indicate where provided: page no/section/legend)	n/a
State number of times the experiment was replicated in laboratory	Cognition was assessed by two independent psychometricians who were blinded to participants' interventions, using a tablet-based cognitive test battery called Inbrain Cognitive Screening Test (	
Define whether data describe technical or biological replicates	Not detected.	

Ethics	Yes (indicate where provided: page no/section/legend)	n/a
Studies involving human participants: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Written informed consent was acquired from all participants and the study protocol was approved by the Institutional Review Board of the Samsung Medical Center (IRB No. 2018-02-050) 2.3 Randomization Consecutively recruited participants were randomly assigned to the intervention or control group.	
Studies involving experimental animals: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why.	Not detected.	

Dual Use Research of Concern (DURC)	Yes (indicate where provided: page no/section/legend)	n/a
If study is subject to dual use research of concern, state the authority granting approval and reference number for the regulatory approval	Not currently checked by SciScore	

Analysis

Attrition	Yes (indicate where provided: page no/section/legend)	n/a
State if sample or data point from the analysis is excluded, and whether the criteria for exclusion were determined and specified in advance.	Under the hypothesis that the standard deviation of scores (at baseline or 12 months and in the control or intervention group) was 15 , and the difference between changes of total score (from baseline to 12 month) of the two groups was five , 142 participants were required for each study group with a significance level set at 0.05 and with a power of 80 % to detect a significant difference.	

Statistics	Yes (indicate where provided: page no/section/legend)	n/a
Describe statistical tests used and justify choice of tests.	Not detected.	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
State whether newly created datasets are available, including protocols for access or restriction on access.	Not detected.	
If data are publicly available, provide accession number in repository or DOI or URL.	Not detected.	
If publicly available data are reused, provide accession number in repository or DOI or URL, where possible.	Not detected.	

Code availability	Yes (indicate where provided: page no/section/legend)	n/a
For all newly generated code and software essential for replicating the main findings of the study:		
State whether the code or software is available.	Not detected.	
If code is publicly available, provide accession number in repository, or DOI or URL.	Not detected.	

Analysis

Adherence to community standards	Yes (indicate where provided: page no/section/legend)	n/a
MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives. Journals have their own policy about requiring specific guidelines and recommendations to complement MDAR.		
State if relevant guidelines (eg., ICMJE, MIBBI, ARRIVE) have been followed, and whether a checklist (eg., CONSORT, PRISMA, ARRIVE) is provided with the manuscript.	Not currently checked by SciScore	